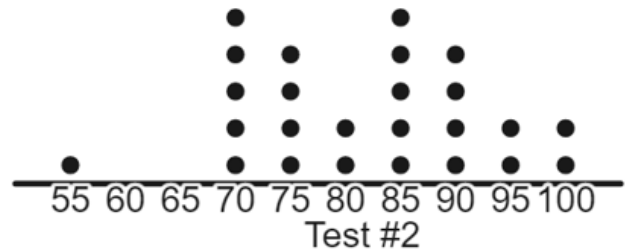
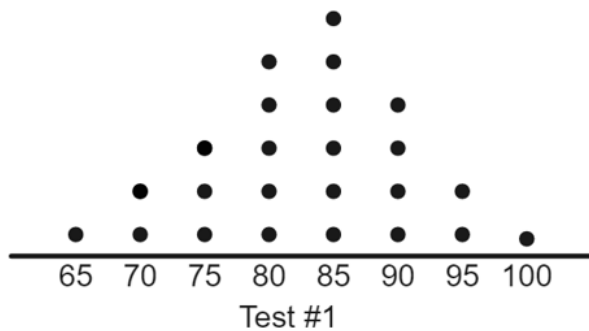


Algebra 1
Unit 12
Lesson 3 Homework

Name Answer Key
Date _____
Period _____

Two dot plots are shown, with results from two recent tests.



1. What is the mean of the scores for Test #1?

$$\frac{1990}{24} \approx 83$$

2. What is the median for the scores of Test #1?

85

3. Which measure of center is the most appropriate to use to describe the scores of Test #1? Explain how you know.

Since the data has a fairly normal distribution, it is most appropriate to use the mean.

4. Find the interquartile range of Test #1 and interpret it in the context of the problem.

$$90 - 77.5 = 12.5$$

The middle 50% of test scores have a range of 12.5.

5. What is the mean of the scores for Test #2?

$$\frac{2040}{25} = 81.6$$

6. What is the median of the scores for Test #2?

85

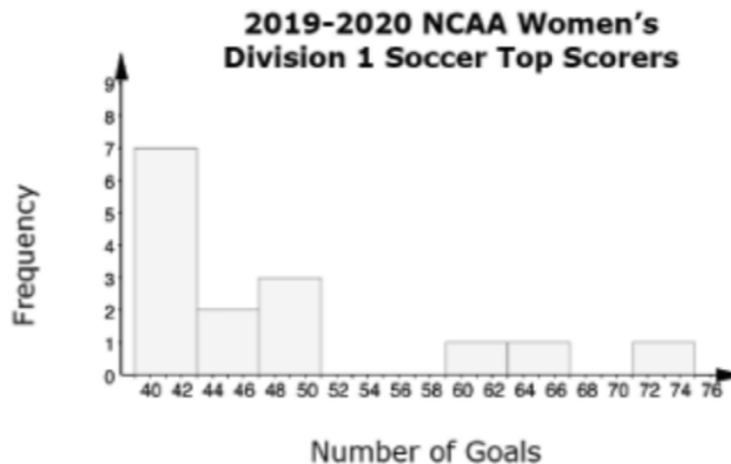
7. Which measure of center is most appropriate to use to describe the scores of Test #2?

Median

8. Write a sentence comparing the results of Test #1 to Test #2 using measures of center and measures of spread.

The results of Test #1 have a fairly normal distribution, so it is best to use mean and standard deviation; the results of Test #2 are skewed, making median and IQR the best choice.

9. In the histogram shown, which measure of center would best be used to describe the data? How do you know?



Median – because the data is skewed

10. A table of values for the heights of the students in a class has a mean of 61 inches and a standard deviation of 1.4 inches. Interpret the variation in height.

A standard deviation of 1.4 is fairly low, meaning the majority of students are close to the average height of 61 in.

The students' heights range from 59.6–62.4 inches.